

Sycophancy Impact Card			
Sharma et al. (2024) — Towards Understanding Sycophancy in Language Models. ICLR 2024.			
Behavior(s) measured	• Opinion conformity: more positive feedback when users claim authorship of content • Position switching: abandoning correct answers when challenged, without new evidence • Answer bias: shifting factual and opinion responses toward users’ stated views • Error mimicry: reproducing users’ mistakes in code and mathematical tasks		
Referent	Position All four behaviors accommodate the user’s expressed or implied position rather than the user themselves. Sub-referents (per Ye et al., 2026) break down as follows:		
	Sub-referent	Behaviors	Rationale
	Explicit	Position switching, answer bias	User directly states a belief or challenges the model’s answer; sycophancy is a response to an overt signal.
	Explicit / Implicit	Opinion conformity	User explicitly claims authorship, but the position (approval-seeking) is inferred; straddles explicit and implicit.
	Implicit	Error mimicry	User’s mistake implies a belief without stating it; model accommodates an inferred rather than declared position.
Norm displaced	Truth / factual accuracy. Models trade correct answers for user approval; accuracy on factual tasks decreases when users signal incorrect beliefs. Calibration is also implicitly displaced (via position-switching under challenge) and <i>evidential independence</i> (via preference model analysis showing reward models incentivize sycophantic responses).		
Outcome measured	• Rate of position switches when the user challenges a correct answer • Change in QA accuracy when users signal incorrect prior beliefs • Frequency with which humans and preference models prefer sycophantic over truthful responses (~20% for human raters) • Accuracy of models trained against sycophancy-prone preference models		
Time horizon	Immediate All outcomes measured at the single-response level. No conversation-level or longitudinal data collected.		
Population	Simulated general users via prompt manipulation (user beliefs implied by prompt framing). No direct human participants in the behavioral studies; crowdworkers used only for preference data analysis. Generalization to real user populations is not established.		
Evidence level	Level 2 — Norm displacement Accuracy is quantified as a displaced norm across behavioral tasks. The preference data analysis (§4) approaches Level 1 (behavioral detection) for the upstream incentive question. No user interaction data; does not reach Level 3.		
Proposed mitigations	• Best-of-N sampling using a non-sycophancy-incentivizing preference model (Claude 2 PM)		

	Upstream identification: analysis of what human preference data incentivizes as a diagnostic tool
Mitigation outcomes	Best-of-N sampling reduces sycophancy for easier misconceptions but remains ineffective for harder ones. Human preference analysis reveals sycophantic responses are preferred a non-negligible fraction of the time, suggesting preference-model-based mitigation has inherent limits tied to the training signal itself. <i>No downstream user impact assessed for mitigations.</i>